Validation & the Full Pipeline

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Where we are

Days 1–2 built a full toolkit:

- **Reduce** — PCA (linear, global variance), UMAP (non-linear, local structure)
- **Cluster** — K-means, GMM, DBSCAN, and “reduce first, then cluster”

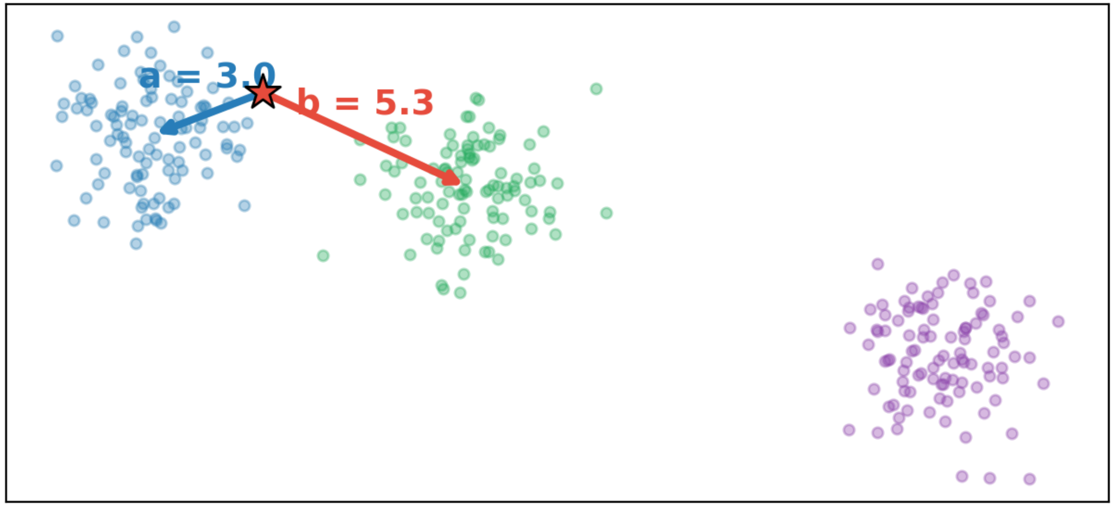
Enough to run a pipeline end to end. But every method so far just *returns* clusters — none of them tells us whether those clusters are real.

We’ve dodged this the whole course:

With no labels, how do we know a clustering is *good*?

The two distances

For a single point, ask two questions: how close is my *own* crowd, and how close is the *nearest other* crowd?



a = average distance to its **own** cluster · b = average distance to the **nearest other** cluster. Good clustering: small a, large b.

The validation problem

Clustering is *unsupervised* — there's no answer key. Any algorithm will happily return clusters, even from pure noise.

Two kinds of check:

- **Internal**: use only the geometry — are clusters tight and well separated? (*silhouette*, inertia, Davies–Bouldin)
- **External**: compare to known labels, *when you happen to have them* (purity, adjusted Rand index)

Internal scores are what you'll usually have. The trap: a good internal score means the *geometry* is clean — not that you found the thing you care about.

Why a new metric?

A good clustering needs *two* things at once:

- **Cohesion** — each point sits close to its own cluster
- **Separation** — clusters are far from each other

Inertia (the elbow metric from K-means) only measures cohesion — and it *always* shrinks as you add clusters. It can't tell you when to stop, and a $K = 3$ score isn't comparable to a $K = 5$ one.

We want one number that rewards cohesion **and** separation, on a **fixed scale**.

The silhouette score

Combine the two distances into one number per point:

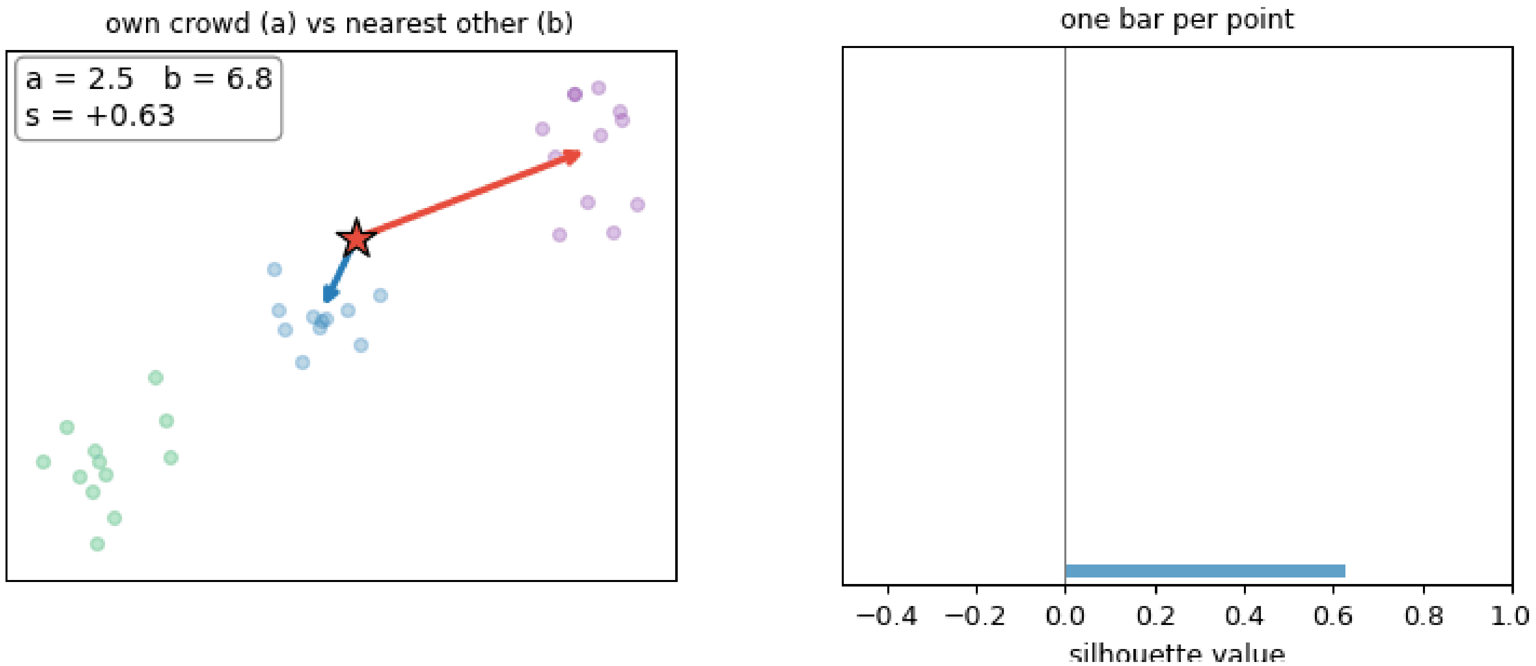
$$s = \frac{b - a}{\max(a, b)} \in [-1, 1]$$

The numerator is the *separation margin* — how much farther the nearest other crowd is than your own. The denominator just rescales it to $[-1, 1]$.

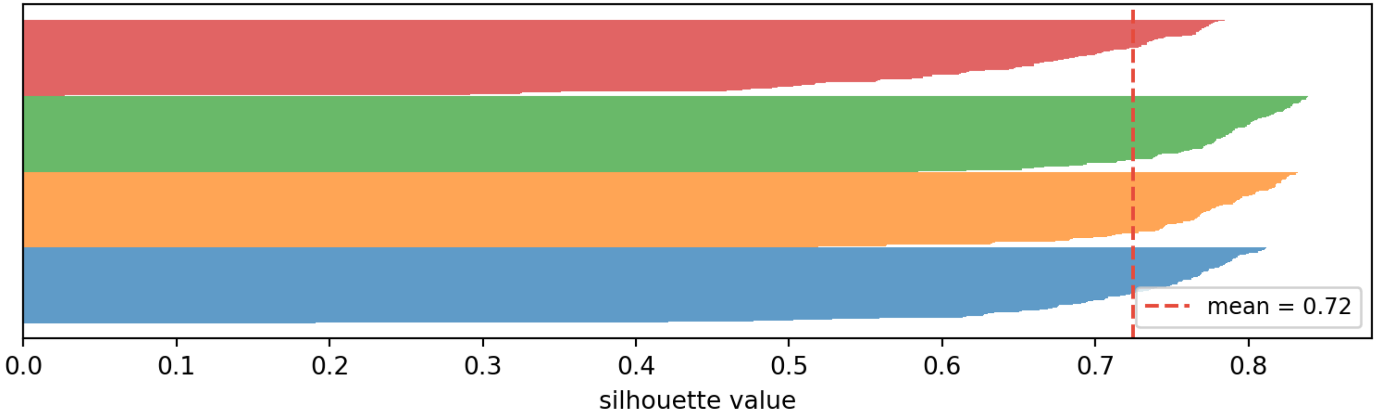
$s \approx 1$: tightly in its own cluster. $s \approx 0$: on a boundary. $s < 0$: probably misassigned.

Building the silhouette plot

Each point contributes one bar. We sweep through the points, compute its a and b, and stack the bars cluster by cluster.



Reading a silhouette plot

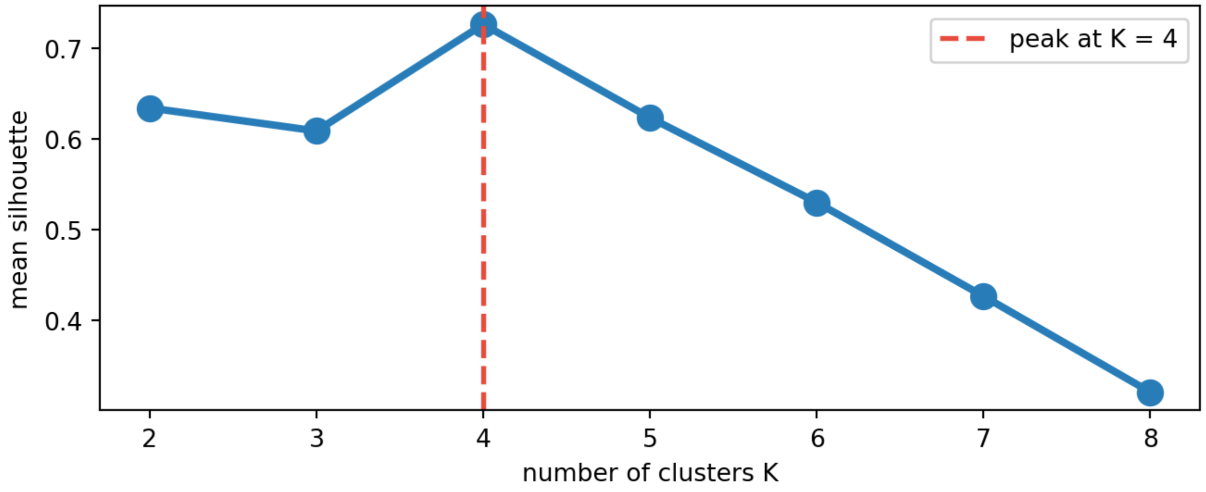


Each bar is one point's s, grouped by cluster and sorted within each — so every cluster shows up as a "blade".

- *Wide, near 1* — the point sits deep in its own cluster, far from any other: a clean, well-separated cluster.
- *Short bars* — the point is near a boundary; the cluster is fuzzy or overlaps a neighbour.
- *Crossing 0 (negative)* — the point is *closer to another cluster* than its own → probably misassigned.

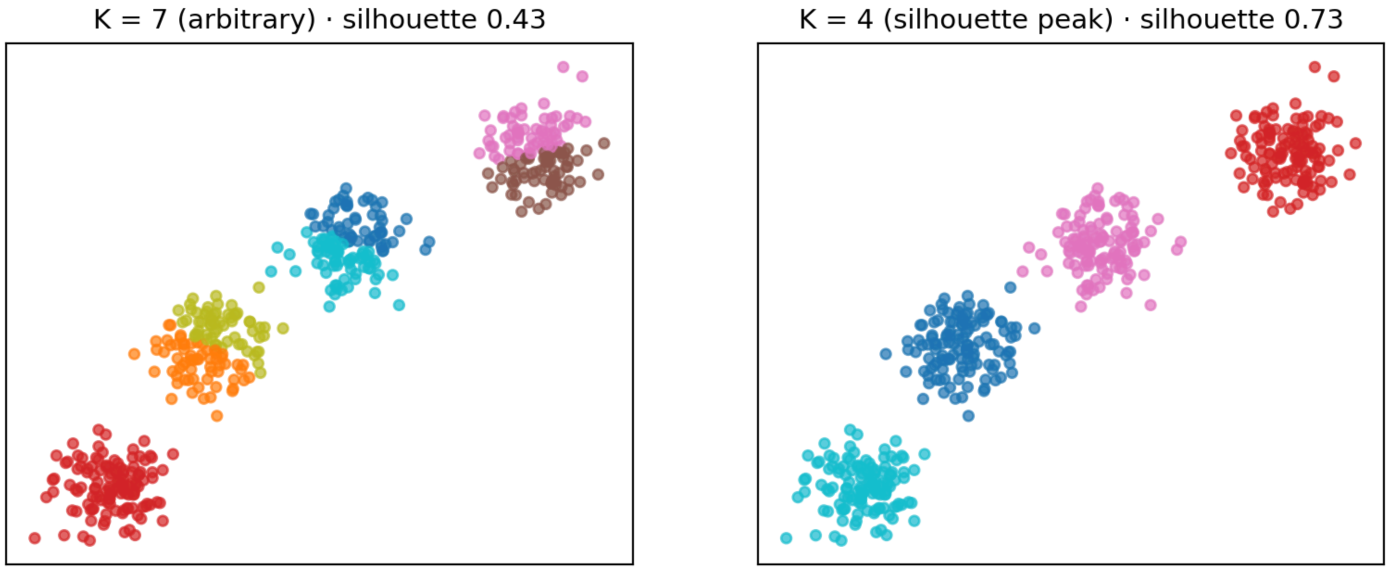
Picking K with the silhouette

Sweep K, take the mean silhouette for each, and let the **peak** suggest how many clusters the geometry supports — no labels needed.



Sharper than the elbow: the score can go *down* when you over-split.

Arbitrary K vs the suggested K



Pick $K = 7$ for no reason and K-means *chops the four real blobs into fragments* — the score drops. The peak K matches the structure and scores highest. **The metric picks the K, not you.**

Three ways to set the knob

The elbow, the k-distance plot, and the silhouette are the **same idea**: try to figure out the tuneable parameter (K or `eps`)

Tool	Sets	For	Scores	Watch out
Elbow	K	K-means / GMM	inertia — *cohesion only*	no true minimum; eyeball the "elbow"
k-distance	`eps`	DBSCAN	local density gap	needs a sparse background
Silhouette	K / any labels	most methods	cohesion **+** separation	misjudges non-convex shapes

Two of them answer *"how many clusters?"* (elbow, silhouette); the k-distance plot sets a *radius* — DBSCAN finds the count itself. The silhouette is the most principled (bounded, with a real peak), which is why it can over-rule the elbow — but **no score replaces checking the science**

The pipeline, end to end

Everything from Days 1–2, as one loop:

1. ***Scale*** — so distance isn't dominated by units
2. ***Reduce*** — PCA / UMAP to kill noise dimensions
3. ***Cluster*** — K-means / GMM / DBSCAN
4. ***Validate*** — silhouette *and* a check against known science

It's a **loop, not a line**: a bad score sends you back to re-scale, re-reduce, or switch algorithm.

Other methods you'll meet

One you'll see in papers but we won't drill into:

- *t-SNE* — a non-linear embedding like UMAP, but it distorts cluster sizes and between-cluster distances; reach for UMAP instead, and never read distances off a t-SNE plot.

Same playbook regardless: reduce if needed, cluster, then *validate*.

Summary

- How do we know if clustering is good without labels?
- **Internal validation** (silhouette) scores *geometry* — necessary, not sufficient
- **External validation** (purity, completeness) when you have any labels
- **The pipeline is a loop**: scale → reduce → cluster → validate → repeat
- **Notebook** `10` — full pipeline on real pulsar data; finish Part 5

Appendix: PCA vs t-SNE vs UMAP

For reference when you meet these in papers:

	PCA	t-SNE	UMAP
Preserves	global variance	local neighbours	local + some global
Linear?	yes	no	no
New points?	yes	no	yes
Distances mean something?	yes	no	roughly

Default to PCA to *understand* variance, UMAP to *see* non-linear structure; treat t-SNE as view-only.